

Heritability of Response Inhibition in Children

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Abstract

We report the heritability of response inhibition, latency, and variability, which are potential markers of genetic risk in neuropsychiatric conditions. Genetic and environmental influences on cancellation and restraint, response latency, and variability measured in a novel variant of the stop signal task were studied in 139 eight-year-old twin pairs from a birth cohort. Cancellation (50%), restraint (27%), and response latency (41%) showed significant heritability, the balance being non-shared environmental influences and/or error. Response variability was not heritable, with 23% of the variance attributable to shared environmental influences and 77% to non-shared environmental risk or error. The phenotypic correlation between response cancellation and restraint was $-.44$ and between response latency and restraint was $.21$. These phenotypic correlations were entirely genetic in origin. The phenotypic correlation between response variability and % successful inhibition was $.27$, but was not genetic. Cancellation and restraint were heritable and shared genetic influences, indicating that they may be influenced by a common gene or genes. Response latency was moderately heritable and shared genetic influences with restraint, but was not correlated with cancellation. Response variability was not heritable. These results support the potential of response inhibition and latency as endophenotypes in genetic research. (*JINS*, 2011, *17*, 238–247)

Keywords: Twin modeling, Endophenotype, Stop task, Go/no-go, Inhibition, Heritability

INTRODUCTION

In an effort to increase the power of genetic association studies of psychiatric disorders, there is growing interest in the potential of intermediate phenotypes, often called endophenotypes (Almasy & Blangero, 2001; Crosbie, Perusse, Barr, & Schachar, 2008; Gottesman & Gould, 2003). Endophenotypes are traits which reflect genetically mediated variation in the pathways linking genes to disease phenotypes. They are presumed to be closer to the action of genes than is the disease phenotype. Disease phenotype refers to those symptoms of the disease that are directly observable; endophenotypes are often less directly observable and may require special methods such as neuropsychological testing or imaging for measurement. Endophenotypes could be powerful tools in genetic research if they can identify a subset of disease-causing genes, point to genetically informative subgroups of affected individuals, or delineate genetically vulnerable

individuals. Quantitative endophenotypes avoid assumptions about the appropriate threshold for disease diagnosis that are embedded in existing diagnostic algorithms (Lee et al., 2008), permit the use of statistically powerful parametric analyses in genetic research, and inform the mechanism by which genetic risk, once identified, is translated into behavior (Merikangas & Risch, 2003).

To be genetically informative, a valid endophenotype must distinguish individuals with a disease diagnosis from those without, differentiate those who are at genetic risk for disease such as unaffected relatives of individuals with the disorder from those who are not, and segregate in families with the phenotype. Although any measure could function as an endophenotype, it is advantageous if it is feasible given that genetic research is likely to require the study of a large number of participants often across multiple sites. Finally and critically, the endophenotype should be heritable. It is possible that there are multiple individual endophenotypes for any complex disorder, each one marking a subset of genetic risks and a unique pathway from genetic cause to phenotype. It is also possible that some endophenotypes are genetically related because they reflect common cognitive and neurobiological mechanisms.

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In these cases, combinations of endophenotypes could prove to be more genetically informative than any single measure in isolation.

Much research has focused on the role of executive functions as potential endophenotypes of neuropsychiatric disorders. Executive functions refer to a range of cognitive control capacities such as planning, performance monitoring, working memory, and response inhibition (Norman & Shallice, 1986). Executive functions play a crucial role in scheduling and optimizing subsidiary processes that are required to meet changing goals in a flexible manner (Mesulam, 1998). Executive abilities depend upon partially overlapping and partially distinct neural, neurochemical, and genetic networks (Shallice & Burgess, 1996). They play a critical role in normal development (Zelazo, Carter, Reznick, & Frye, 1997), in behavior (Logan, Schachar, & Tannock, 1997; Young et al., 2009), and in various psychopathologies (Cannon & Keller, 2006; Crosbie et al., 2008). Among the most studied of executive control endophenotypes in child psychopathology are response inhibition (the ability to stop a pre-potent response), response latency (the ability to prepare and execute a speeded motor response), and response variability (the ability to respond in a consistent manner) (Sergeant, 2000). Response inhibition has been parsed into two major components: restraint and cancellation. Restraint comes into play when prepared responses are withheld pending further information such as a possible signal to stop a pending action; cancellation refers to withdrawal of an ongoing response. Restraint is akin to slowing down while driving in a residential neighborhood; cancellation is analogous to slamming on the brakes when someone runs onto the road. Restraint and cancellation are partially distinct, both anatomically (Band & Van Boxtel, 1999; Chambers et al., 2007; Chevrier, Noseworthy, & Schachar, 2007) and behaviorally (Schachar et al., 2007), and may involve frontal-subcortical networks (inferior prefrontal cortex, basal ganglia, and midbrain nuclei). They play a crucial role in movement (Band & Van Boxtel, 1999), normal development (Harnishfeger & Pope, 1996; Williams, Ponesse, Schachar, Logan, & Tannock, 1999), various neuropathologies, for example, Parkinson's disease and brain injury (Gauggel, Rieger, & Feghoff, 2004; Seiss & Praamstra, 2004), and several psychiatric disorders (e.g., attention deficit hyperactivity disorder [ADHD], schizophrenia, and obsessive-compulsive disorder [OCD]; Lipszyc & Schachar, 2010; Crosbie et al., 2008; Rommelse, Arias-Vasquez et al., 2008).

Typically, cancellation is measured in the stop signal task (SST) and restraint in the go/no-go task (GNG). In both tasks, participants are asked to respond as quickly as possible on most trials, but to stop their responses if they can on a subset of trials when a stop signal is presented. However, the GNG and SST tasks differ in a critical way. In the GNG, the delay between presentation of the go and the stop signals is zero. Consequently, the GNG demands inhibition during response preparation. Accordingly, in the GNG task, the primary index of the ability to restrain a response is the probability of successful inhibition (or its inverse commission errors). In the SST, the stop signal follows the presentation of the go signal and is adjusted dynamically depending on whether a response

is inhibited. As a result, the response will be under way when the stop signal appears. Therefore, the SST demands cancellation of an ongoing response. If a response is successfully inhibited, the delay is increased on the next stop trial, thereby increasing difficulty. If the participant fails to inhibit their response, the delay is shortened making it easier to inhibit on the next trial. Whether or not a response can be stopped is modeled as a function of the outcome of the race between response execution and inhibition processes; whichever process finishes first determines whether or not a response will be stopped. Given the assumptions of the race model (Verbruggen & Logan, 2008b), the latency of the inhibition process (known as stop signal reaction time, SSRT), which is otherwise not directly observable, can be estimated. For the current study, we developed a hybrid of the SST and GNG tasks in which half the stop signals appear after some delay as in the typical SST and half appear after zero delay as in the GNG task.

Family studies (Rommelse, Altink, Martin et al., 2008; Schachar et al., 2005; Uebel et al., 2010) and a twin study (Friedman et al., 2008; Young et al., 2009) of cancellation indicate that it is a heritable trait (.48) which shares most of its genetic influence with performance on other measures of inhibition (e.g., antisaccade and Stroop tasks) and with disinhibited behavior. Kuntsi et al. (2006) estimated the heritability of restraint (commission errors) in the GNG task to be between 18 and 45%.

Longer and more variable responses are characteristic of various neuropathologies (Hetherington, Stuss, & Finlayson, 1996) and psychopathologies such as ADHD (Bidwell, Willcutt, Defries, & Pennington, 2007), autism (Townsend, Harris, & Courchesne, 1996), anxiety disorder, schizophrenia, and reading disability (Lipszyc & Schachar, 2010). Twin studies, largely in adults, indicate genetic influences on reaction time and reaction time variability ranging from 30 to 50% (Finkel & McGue, 1998; Kuntsi et al., 2006; Posthuma, Mulder, Boomsma, & de Geus, 2002; Simonen, Videman, Battie, & Gibbons, 1998). Others have reported essentially no evidence of heritability for response variability (Finkel & McGue, 1998, 2007; Groot, de Sonneville, Stins, & Boomsma, 2004; Sanders et al., 2006; Stins et al., 2005). Heritability of response latency and variability as measured in the stop task has not been reported.

Although supportive of the conclusion that restraint and cancellation are heritable, these studies leave several issues unresolved. The study by Friedman et al. (2008) used a non-standard version of the stop task in which words were categorized as animal or non-animal, a variation that could affect performance. Moreover, the average age of participants in this study was 17 years, which is substantially older than is typical of children with most childhood psychiatric disorders such as ADHD. Finally, there has been no study of whether restraint and cancellation share genetic risk and whether response latency or variability measured in these tasks are heritable. Accordingly, the primary goal of this study was to apply behavioral genetic methods to estimate the genetic and environmental influences on inter-individual variability in two variants of response inhibition (restraint and cancellation),

response latency and response variability. Given their possible cognitive and anatomical overlap, the study used multivariate methods to estimate the extent to which these executive abilities share genetic and environmental substrates. If executive function abilities have common genetic influences, they could be used individually or combined as endophenotypes in genetic research. The study used a task that incorporated stop signal and GNG trials in a single task to measure restraint and cancellation. We studied a cohort of 100-month-old twins to determine if these executive function measures are heritable in school-age children.

METHOD

Participants

The Quebec Newborn Twin Study (QNTS) is based on a representative sample of twins born between April 1995 and December 1998 in the Greater Montreal Area, Canada. Names, addresses, and phone numbers of all mothers of newborn twins were collected from the birth records of the Quebec Bureau of Statistics. Participating families were assessed yearly, starting when the children were 6 months old corrected for gestational age. Families were predominantly of European descent (84%), with the remainder of African (3%), Asian (2%), and other (2%) descent. Nine percent did not provide information on ethnicity. Severe neonatal and perinatal complications precluding participation in the 5 month assessment or an inability to communicate in either French or English as assessed during the initial telephone contact were the only exclusion criteria.

The maternal language of the parents was French for 71.9% of mothers and 72.3% of fathers, English for 9.9% of mothers and 8.8% of fathers, and another language for 19.2% of mothers and 18.9% of fathers. The average household income was around CAN \$54,000, which is slightly above average for similar households in this geographic area at that time. Fourteen percent of mothers did not complete high school, 28% were high school graduates, 48% had some college education or graduated from college (9% did not provide any information regarding education). Eleven percent of fathers did not complete high school, 34% were high school graduates, and 39% had some college education or graduated from college (information was unavailable for 16% of fathers).

A total of 989 families were asked to join the QNTS; of these, 662 were assessed during the first wave of data collection. The average attrition rate from 6 to 84 months was $-.05\%$ per year, ranging between $-.16\%$ and $+.20\%$. Only the twins who reached 100 months between April 2004 and October 2005 ($n = 139$ pairs) were assessed for the present study because of funding availability. Of these, the inhibition scores were available in 234 participants after removing extreme scores distributed in 131 pairs (only one twin was reliably tested in 28 pairs) (49.6% boys and 51.4% girls). Of these, 55 pairs (42%) were monozygotic (MZ) and 76 pairs (58%) were dizygotic (DZ). The mean age was 100.7 months

($SD = 1.33$). Data for latency and response variability were available for 267 individual twins (115 MZ and 152 DZ).

The families of the subsample of twins assessed at 100 months differed little from the original sample: the parents of the twins that were evaluated at 100 months were more likely to have French or English as their first language (mothers: $\chi^2 = 16.043$, $df = 1$, $p = .000$; fathers: $\chi^2 = 16.609$, $df = 1$, $p = .000$). Most of the missing data were missing by design. Missing status in participants was associated with one observed socio-demographic characteristic, but there was no reason to think that the missing values on the experimental measures would have been, if known, a non-random subsample of all the values for these measures. For this reason, the missing data were considered missing at random (MAR) (McKnight, McKnight, Sidani, & Figueredo, 2007). To maximize power, all available data were analyzed and the model parameters were estimated with the Full Information Maximum Likelihood (FIML) method. The FIML estimation procedure is implemented directly in the process of fitting a model; it treats missing data by fitting the model to all non-missing data for each observation (McCartney, Burchinal, & Bub, 2006). There is evidence that FIML results in unbiased parameter and standard error estimates (Graham, 2009) and that it has more power than multiple imputation unless a great number of imputations are done (Graham, Olchowski, & Gilreath, 2007). The downside of the FIML procedure is that a single N is not representative of the study sample. All the genetic models were estimated using the FIML method.

Hybrid-Stop Task

Response inhibition, response speed, and response variability were measured using a version of the SST known as the Hybrid Stop Task (HST) (Figure 1). The HST was designed to measure both restraint as in typical GNG tasks and cancellation as in typical SST in a single paradigm. Like the GNG and SST, the HST involved two types of trials (go and stop trials). Blue and black arrows (4×4.5 cm) presented for 500 ms were used as stimuli for the task. In go trials, participants are instructed to respond as quickly and accurately as possible with either a left or a right button press depending on the direction of a black arrow. On 50% of trials and at random, a blue arrow appeared instead of the black arrow instructing participants to stop their response on that trial if they could. There were two kinds of stop trials, no-go (GNG) and stop-signal (SST) trials, indicated by presentation of a blue arrow. In GNG trials, the blue arrow appeared *at the same time* as and instead of the go arrow thereby creating conditions typical of GNG tasks in which there is no delay between presentation of the go and the stop signal. In other words, the go arrow appeared in blue rather than black indicating that the participant should withhold their response on that trial. The primary index of inhibition in no-go trials was the percent of successfully inhibited trials (the inverse of commission errors) (% inhibition). In SST trials, the blue no-go arrow was presented *after* some variable stop-signal

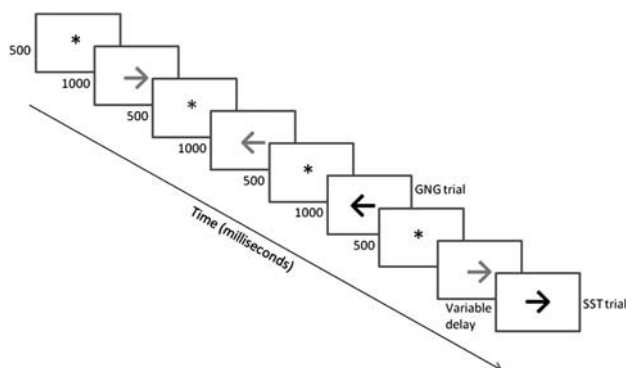


Fig. 1. Hybrid Stop Signal Task (HST). Each trial began with a fixation dot in the middle of the screen for 500 ms followed by a go stimulus. The go stimulus was a blue arrow pointing either to the left (indicating a left hand response) or to the right (indicating a right hand response). The go stimuli remained on screen for 1000 ms; 50% of trials contained a stop signal (presentation of a black rather than a blue arrow). In no-go trials, the stop signal appeared instead of the go arrow. In stop-signal trials, the stop signal always appeared following some delay after the go signal. The stop signal indicated that the subject should not make the given response on that trial. Delay between go and stop signals was adaptively adjusted in stop signal trials to ensure that half of stop signal trials could not be stopped, despite the appearance of the stop signal. A blank screen occupied the remaining time between trials.

delay—the go arrow would appear in black and would change to blue after some variable delay creating a situation in which the subject would have initiated the response when the signal to stop was presented. A dynamic tracking algorithm was used to adjust the delay in SST trials. After an initial delay of 250 ms, stop signal delay was dynamically adjusted in 50 ms steps to converge on the delay at which individuals were able to stop 50% of the time. If the subject could successfully withhold their response, the program increased the delay by 50 ms, rendering the next stop trial more difficult. If the subject was unable to successfully inhibit his (her) responses following a stop signal, the program decreased the delay by 50 ms for the next stop trial, making the next stop trial easier. At the delay which allowed for 50% of trials to be stopped, the outcome of the race between the go response process and the stop process is tied on average allowing for an estimate of the latency of the inhibition process known as stop signal reaction time (SSRT) using the formula ($SSRT = \bar{u} \text{ go RT} - \bar{u} \text{ stop signal delay}$), where mean go reaction time was based on responses in trials when there was no stop signal and stop signal delay was the delay between presentation of go signal and presentation of the stop signal at which the probability of inhibition is .5 (Logan, 1994). We excluded several participants with invalid task participation as is typical of stop task studies (e.g., probability of inhibition greater than 80% or less than 20% or error rates over 60%).

Stimuli were presented on a light gray background of a Dell TM 1800FP color monitor. Participants were instructed to respond with the hand indicated by the direction of the

black arrow and to squeeze a strength gauge as quickly as possible in response to an arrow for the go trials (respond with the right hand to a right-pointing arrow and with the left hand to a left-pointing arrow), but to withhold the action for the no-go and stop signal trials (blue arrow). There were 64 trials per block and 5 blocks in total. Among the 320 trials, 50% were go trials ($n = 160$), 25% were no-go trials ($n = 80$), and 25% were stop-signal trials ($n = 80$). Between blocks, 20-s pauses were interspaced. Both the colors of arrows and the stimulus orders were randomized (i.e., for half the children, black indicated go while blue indicated no-go/stop-signal; for the other half, blue indicated go while black indicated no-go/stop signal trials). Presentation of arrow lasted for 500 ms, each trial lasted for 3 s, and the task was 18 min in duration. Soreni et al. (2009) reported high test-retest reliability of SSRT (intra-class correlation [ICC] = 0.72), go RT (ICC = 0.62), and go RT SD (ICC = 0.74) over three test administrations separated by 1 week each. The study procedure was approved by the Institutions Research Ethics Board.

Analyses

We examined the common genetic and environmental etiology of response inhibition (in GNG and SST trials), go RT, and go RT SD using classic twin modeling estimated by structural equation modeling and decomposed the covariance of correlated measures into genetic and environmental components. This design is based on the comparison of the covariance between MZ and DZ twins for a given measure knowing that the genetic correlation is 1 between MZ twins and .5 on average between DZ twins. Formal analysis of twin data was conducted using the Mx package (Neale, Boker, Xie, & Maes, 1999) to partition the variance of measures between additive genetic (A) and environmental sources (Plomin & Kosslyn, 2001). The environmental variance was further decomposed to estimate the proportion of variance attributable to the common environment (C), which augments the similarity between twins regardless of zygosity, and to the unique environment (E), which makes twins of a pair more different (Turkheimer & Waldron, 2000). For % inhibition in GNG trials, we applied a reflect-and-log transformation to correct for skewness and another reflect to restore the direction of the scale (Tabachnick & Fidell, 1996, 2007). We estimated multivariate models for pairs of variables that were significantly correlated; where there is no phenotypic correlation, genetic and environmental correlation would have to be non-significant as well.

RESULTS

Descriptive Statistics

The means and standard deviations for response inhibition, variability, and latency are reported in Table 1 for all participants by zygosity type. There were no significant mean and variance differences between MZ and DZ twins. There was a significant

Table 1. Descriptive statistics

	All participants		MZ twins			DZ twins		
	Mean	SD	Mean	SD	ICC	Mean	SD	ICC
Go RT	698.96	97.69	701.03	100.83	.43	697.34	95.58	.20
Go RT SD	213.65	44.33	212.92	44.46	.24	214.21	44.39	.22
SSRT	237.44	170.40	250.25	157.37	.45	227.76	179.54	.26
% inhibition	88.83	11.40	89.80	9.54	.35 ^a	88.10	12.60	-.02 ^a

Note. MZ = monozygotic twins; DZ = dizygotic twins; ICC = intra-class correlation; Go RT = go reaction time; Go RT SD = standard deviation of go reaction time; SSRT = stop signal reaction time; % inhibition = % successful inhibition; ^aICC for % inhibition based on transformed variables.

mean difference in go RT between boys and girls (boys mean = 676.9 ms, girls mean = 720.6 ms, $t(232) = -3.50$, $p = .001$). Age was unrelated to the outcomes (the correlations ranged between $r = -.05$ and $r = .06$, all non-significant).

Genetic Modeling

The difference between the intra-class correlations of MZ and DZ twins reported in Table 1 suggests that response inhibition (both % inhibition in GNG and SSRT in SST trials) and go RT were substantially heritable, but that go RT SD was not. The full ACE model (additive genes and common and unique environment model) showed an adequate fit when compared with a saturated model in all cases (SSRT: $-2LL$ ratio = .028, $df = 1$, $p = .64$; go RT SD: $-2LL$ ratio = .000, $df = 1$, $p = 1.0$; go RT: $-2LL$ ratio = 2.04, $df = 1$, $p = .153$; % successful inhibition. We also tested three nested models: the AE model (additive genes and unique environment), CE

model (common and unique environment), and the null E model (unique environment only).

The results of the univariate genetic models and nested models for % inhibition, SSRT, go RT SD, and go RT are reported in Table 2. In the case of SSRT, % inhibition and go RT, the common environment component could be removed without deteriorating the fit of the model compared with the full ACE model. The null model was rejected in the case of go RT and SSRT and close to rejection in the case of go RT SD and % inhibition. Although the CE model was not always formally rejected, probably because of low power, it showed more fit deterioration than the AE model when compared with the ACE model. Together, these results show that the AE model was the best and most parsimonious representation of the genetic and environmental etiology of the % inhibition, SSRT, and go RT.

There was a significant correlation between % inhibition in GNG trials and SSRT in SST trials ($-.41$) (see Table 3).

Table 2. ACE and nested models for stop task outcomes

Model	χ^2	df	p	a^2 (95% CI)	c^2 (95% CI)	e^2 (95% CI)
Go RT						
ACE	—	—	—	.41 (.00–.59)	.00 (.00–.44)	.59 (.41–.82)
AE	.00	1	1.00	.41 (.20–.59)	—	.59 (.41–.80)
CE	1.58	1	.210	—	.32 (.14–.47)	.68 (.63–.86)
E	13.08	2	.001	—	—	1
Go RT SD						
ACE	—	—	—	.02 (.00–.47)	.21 (.00–.40)	.76 (.53–.96)
AE	.56	1	.46	.28 (.05–.48)	—	.72 (.52–.96)
CE	.01	1	.95	—	.23 (.04–.40)	.77 (.60–.96)
E	5.79	2	.06	—	—	1
SSRT						
ACE	—	—	—	.50 (.00–.66)	0 (.00–.40)	.50 (.34–.73)
AE	0	1	1.00	.50 (.29–.66)	—	.50 (.34–.71)
CE	3.25	1	.07	—	.34 (.17–.48)	.66 (.51–.83)
E	17.76	2	.000	—	—	1
% inhibition						
ACE	—	—	—	.27 (.00–.48)	0 (.00–.23)	.73 (.51–.97)
AE	0	1	1.00	.27 (.03–.48)	—	.73 (.52–.97)
CE	2.51	1	.11	—	.13 (.00–.30)	.87 (.70–1)
E	4.80	2	.09	—	—	1

Note. Best fitting model is shown in bold. ACE = additive genes and common and unique environment model; CI = confidence interval; AE = additive genes and unique environment model; CE = common and unique environment model; E = unique environment only model; Go RT = go reaction time; Go RT SD = standard deviation of go reaction time; SSRT = stop signal reaction time; % inhibition = % successful inhibition.

Table 3. Phenotypic correlations

	Go RT	Go RT SD	SSRT	% inhibition
Go RT	1			
Go RT SD	.66**	1		
SSRT	-.10	.05	1	
% inhibition	.22**	-.23**	-.41**	1

Note. Go RT = go reaction time; Go RT SD = standard deviation of go reaction time; SSRT = stop signal reaction time; % inhibition = % successful inhibition.

** $p < .01$; $N = 234$ except for the correlation between % inhibition and SSRT were $N = 267$.

The best fitting bivariate model showed that the correlation between these two forms of inhibition was essentially genetic ($R_g = -1$); C and E could be omitted from the model without a drop in fit. The correlation between mean go RT and SSRT was $-.10$ and was too low to allow estimation of a model (all models included 0 in the confidence interval). The phenotypic correlation between go RT and % inhibition in GNG trials was $.22$ and could entirely be attributed to common genetic substrates ($R_g = .58$); C and E correlations could be omitted from the model without a drop in model fit.

Given that we found a significant difference of mean between boys and girls for the inhibition measures, we also tested a univariate genetic model with sex as a covariate for this outcome. The results were very similar to the simple ACE model ($a^2 = .38$, $c^2 = .00$, $e^2 = .62$). The sample size was too limited to test for sex-limited models where parameters are left to vary across sexes.

The ICC of go RT SD for MZ and DZ twins showed little family aggregation. Although the AE model was not formally rejected, the CE model showed the smallest deterioration of fit compared with the full ACE model. The null model was very close to rejection with a p of $.06$. We conclude from these results that, given the low power and small family aggregation, the CE model was the best and most parsimonious representation of the components of the variance in response variability. Go RT SD was correlated with % inhibition ($-.23$, $p < .0001$), but not with SSRT ($-.01$). Given that the etiology of go RT variability was CE and that of % inhibition was AE, the correlation could only be through E, which could reflect correlated error for these two measures.

The correlation between % inhibition and Go RT and Go RT SD were weak at $.22$ and $-.23$. With the relatively low power afforded by our sample, an attempt at modeling a three-variable model was inconclusive: we could not distinguish between the C and E effects and correlations.

DISCUSSION

The goal of this study was to determine the extent to which individual differences in several executive functions were influenced by genetic factors. We focused on two components of motor response inhibition (restraint and cancellation),

and on response latency and response variability because these indices of executive function are considered to be prime candidates for endophenotypes for several neuropsychiatric disorders. However, data have been lacking on whether or not these traits are influenced by common or specific genetic factors and on whether they are heritable in school-age children. Evidence that these executive functions are heritable is critical to support their utility in the search for the genetic underpinnings of cognition in general and of those psychopathologies such as ADHD, OCD, and schizophrenia that are associated with deficits in these abilities. We also wanted to determine if executive functions share genetic substrates to elucidate the genetics of executive processes.

Using an HST designed to measure response inhibition (restraint and cancellation) as well as response latency and variability in school-age children, we demonstrated that there are substantial genetic influences on inhibition and response latency. Approximately 50% of the variance in cancellation (stop signal reaction time), 40% of the variance in latency of the response, and 27% of the variance in restraint (% successful inhibition) was found to be genetic. By contrast, none of the variance in responses variability was genetic. The balance of the variance in inhibition (cancellation and restraint) and response latency was accounted for by non-shared environmental influences and possibly by error. Response variability was affected by shared environmental influences (23%) and by non-shared environmental risk or error (77%). Restraint as measured in GNG trials and cancellation as measured in the SST trials as well as restraint and response latency largely share their genetic origins. By contrast, we could not fit a model to the correlation between response cancellation (SSRT) and response latency and could not conclude that the two traits shared genetic influence. These results indicate that genetic factors influence some executive functions, in particular response inhibition and latency and, to a lesser extent, restraint. They also shed light on the biology of executive functions, indicating that there are common and separable genetic influences on aspects of executive functions.

The current estimate of heritability of response cancellation in the stop trials (SSRT) is quite consistent with previous results [.48; (Friedman et al., 2008)], even though samples differed in age (adolescents vs. school-age children) and the studies used different versions of the stop task. The observed correlation between restraint and cancellation ($-.41$) is consistent with previous cognitive (Verbruggen & Logan, 2008b) and neuroimaging (Rubia et al., 2001) research which showed that these two forms of inhibition may share cognitive processes, such as response-strategy adjustments on a trial-by-trial basis, priming of task goals, or the formation of stimulus-stop associations (Verbruggen & Logan, 2008a; Verbruggen, Schneider, & Logan, 2008), and neural processes, such as development or function of prefrontal cortices, and fronto-subcortical networks upon which these processes depend (Aron & Poldrack, 2005, 2006; Aron, Robbins, & Poldrack, 2004; Chambers et al., 2006, 2007; Lenroot et al., 2009; Shaw et al., 2007, 2008; van den Wildenberg et al., 2010).

Similarly, we observed a significant correlation between response latency and % response inhibition (.21), which indicates that participants with slower responses can inhibit their responses in GNG trials more easily than those with faster responses. This correlation was essentially explained by genetic influences indicating that restraint inhibition and response latency may also involve common processes. Both processes are active during go phases of tasks like the stop task, that is, while preparing and simultaneously withholding a response.

Previous studies found evidence that response variability is heritable when task pacing was rapid rather than slow, but not all circumstances such (Finkel & McGue, 1998; Kuntsi et al., 2006; Posthuma et al., 2002; Stins et al., 2005). The current study did not find response variability to be heritable (see also Finkel & McGue, 2007). The stop task differs in several critical ways from typical reaction time measures: dynamic tracking of the delay between presentation of the go and the stop signals may have constrained variability in go reaction time and influenced heritability. However, stop trials constitute only a minority of total trials in the HST and other stop task variants. Even though increased response variability is characteristic of several psychopathological conditions (Bidwell et al., 2007; Geurts et al., 2008; Johnson et al., 2007; Leth-Steensen, Elbaz, & Douglas, 2000; Rommelse, Altink, Oosterlaan, et al., 2008; Vaurio, Simmonds, & Mostofsky, 2009), the current results counsel caution in concluding that response variation is heritable and, therefore, suitable as an endophenotype for genetic research. At the very least, the current results suggest that variability may itself be variable across tasks, with some indices of variability being more heritable than others. The matter would be worthy of further research.

Along with previous research (Friedman et al., 2008), the current evidence indicates that there is both genetic diversity and genetic uniformity among executive functions. At one level, there are genetic influences that affect all executive functions, for example, inhibition and processing speed. At other levels, there are genetic influences on clusters of processes such as inhibition and others that act on single measures such as restraint and on cancellation (Miyake et al., 2000; Young et al., 2009). Friedman et al. (2008) estimated the heritability of a latent variable of inhibition derived from performance on the stop task, the Stroop task, and an antisaccade task and found it to be nearly 100% heritable. As we did not have a third measure of response inhibition, we did not estimate a latent inhibition trait. However, we note that the correlation between restraint and cancellation inhibition observed in the current experiment ($-.41$) was greater than the correlation among the three inhibition measures in the Friedman et al. (2008) study (.31 for SSRT in SST and antisaccade task; .18 for SST and Stroop task), suggesting that inhibition measures are very likely to contribute to a latent inhibition construct. Processing speed as indexed by response latency appears to relate to more specific executive functions such as inhibition at the highest level of the executive function construct.

There are several potential limitations to this study, including the relatively small sample, single measurement occasion, and the possibility that shared genetic influences on restraint and cancellation were a function of the fact that these indices were derived from a single task. To assess the generality of the HST results, we tested a separate sample of 128 school-aged participants (58 with ADHD; 69 healthy controls) on three measures, that is, the HST, the GNG and the SST tasks, and found very high and significant correlation among all performance indices across tasks. Nevertheless, it would be helpful to confirm the current conclusions in a study which uses separate measures of restraint and cancellation. Low power and absence of phenotypic correlation among several measures precluded testing of genetic models for all four cognitive measures.

In conclusion, the current results add to the evidence about the validity of inhibition as a potential endophenotype by confirming that there are substantial genetic influences on response cancellation and on response restraint (Kuntsi et al., 2006; Rommelse, Altink, Martin et al., 2008; Young et al., 2009). These increasingly validated putative endophenotypes could be used for the selection of individuals at the extremes of the trait ("extreme trait" or "selective genotyping")—a particularly powerful design for detecting genetic association (Huang & Lin, 2007; Van Gestel, Houwing-Duistermaat, Adolfsson, van Duijn, & Van Broeckhoven, 2000). The approach has been successful in studies of cardiac disease and QT interval (Arking et al., 2006), obesity (Herbert et al., 2006), and fetal hemoglobin level (Menzel & Thein, 2009). Genotyping only a subset of extreme trait individuals is also cost-effective compared with genotyping all individuals in a large population sample. There are several other advantages of valid neuropsychological endophenotypes in genetic research. First, there is increased ease of collection of endophenotypes due to the greater availability of general population samples compared with clinically ascertained subjects. Second, neuropsychological measures such as the SST could prove useful in multisite studies given the ease with which they can be standardized in comparison to behavioral measures. And third, neuropsychological task performance can be linked directly to brain function via a range of imaging strategies.

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